FABRID-IDS: Federated Alert-Budget Reallocation for Heterogeneous IoT Intrusion Detection

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Abstract

A frozen federated anomaly scorer can rank attacks well while still producing unreliable operating thresholds across heterogeneous clients. FABRID-IDS treats post-training threshold selection as federated alert-budget reallocation: client-specific nominal false-positive-rate (FPR) targets are assigned under a shared weighted budget and then calibrated independently on local benign data. The primary evaluation uses nine N-BaIoT devices, ten detector seeds, and five nominal budgets; CIC IoT-DIAD provides a secondary protocol stress test with different clients, features, and a different scorer. At $\boldsymbol{B = 0.001}$, unrestricted Macro and Tail allocations realize a federation FPR of **0.00170**, or $\mathbf{1.696}\boldsymbol{B}$, despite satisfying the nominal allocation constraint. Conservative calibration reduces the realized FPR to **0.00036** but lowers macro recall from **0.3778** to **0.2445** and does not restore worst-client recall. Equal-FPR achieves higher macro recall at several nominal budgets, although its realized FPR also exceeds the budget at several operating points. Mean AUROC and AUPRC are **0.9995** and **0.9999**, yet FABRID-family minimum-client recall remains zero across the evaluated N-BaIoT budgets. Thus, satisfying the nominal federation-level alert-budget constraint does not guarantee the realized FPR or client-level detection behavior after finite local calibration, despite near-perfect pooled ranking metrics.

Keywords: federated intrusion detection, Internet of Things, alert-budget reallocation, false-positive-rate constraint, threshold calibration

1 Introduction

Federated IoT detection distributes model training across clients, but deployment still requires a threshold that converts anomaly scores into alerts [1–3]. Shared and client-specific thresholding have been studied [4–6], yet neither determines how a federation-wide FPR budget should be divided across heterogeneous clients. With a frozen scorer, differences in benign-score distributions and attack-recall curves can produce uneven false-alarm and detection-utility outcomes across clients.

FABRID-IDS treats this as alert-budget reallocation. Given a fixed scorer, it uses labeled validation attacks to select client-specific nominal rates α_k under the federation constraint $\sum_k w_k \alpha_k \leq B$. Independent local benign calibration sets the thresholds used for held-out evaluation, allowing us to compare the allocated targets with the observed operating points. B is a nominal federation-level FPR budget, not a realized-FPR guarantee.

The contributions are federation-constrained alert-budget allocation, a frozen-scorer protocol separating allocation, calibration, and testing, and paired N-BaIoT evidence that nominal compliance need not preserve realized FPR or worst-client recall.

2 Related work and scope

Federated IoT IDS research has mainly focused on detector training [1–3, 7], while threshold regulation and distributed anomaly detection address related operating-point questions [8, 9]. Laridi et al. [4] aggregate client summary statistics into one global anomaly threshold. Asiri et al. [5] compute local 95th-percentile benign reconstruction-error thresholds and average them into one global threshold. Fed-DTCN [6] uses held-out benign calibration data to select client-specific thresholds for target local FPRs.

Neyman-Pearson methods address explicit error-rate control [10], while conformal calibration provides finite-sample control under stated sampling or exchangeability assumptions [11]. Recent federated conformal work also examines how trained predictors and calibration-score aggregation affect prediction intervals [12]. Unlike the threshold methods above, FABRID-IDS jointly selects client-specific nominal rates under $\sum_k w_k \alpha_k \leq B$, recalibrates thresholds on disjoint benign data, and evaluates realized federation and client-level FPR. The contribution lies in this federation-level allocation; lower-tail objectives, order statistics, mixed-integer optimization, and finite-sample calibration are established components. The formulation does not guarantee control of the realized federation FPR after local calibration. N-BaIoT's released source-order splits establish neither deployment chronology nor exchangeability.

Training-time allocation is related but addresses a different decision. Fair federated learning and agnostic federated learning vary how model training responds to client performance or mixtures of clients [7]. Here the detector is already fixed. Changing α_k changes only the operating threshold used for client k; it cannot improve the ordering of benign and attack scores. This separation lets the same scores support paired threshold-policy comparisons, with scorer quality a shared limitation.

Global-threshold methods also differ from client-specific rate allocation. A common absolute threshold treats score scales as directly comparable; a common nominal FPR asks each client to calibrate its own threshold at the same target rate; FABRID-IDS

lets the target rates vary while constraining their weighted sum. None guarantees a held-out FPR solely through a nominal constraint, and the question tested here is whether reallocating targets improves the observed operating points under the available calibration data.

3 FABRID-IDS

3.1 Protocol and utility

For K clients, a frozen scorer $s_k(\cdot)$ produces anomaly scores. Released source order defines benign training (50%), frontier (20%), final-calibration (10%), and test (20%) splits; the initial 20% of each retained attack subtype forms the validation split and the remainder forms the test split. The pipeline proceeds through training, frozen scoring, frontier utility estimation, allocation, final calibration, and testing; final-calibration and test benign records are not allocation inputs.

For ascending benign scores $x_{(1)}, \ldots, x_{(n)}$ and nominal target α, threshold calibration uses a standard order-statistic construction [11]

$$\tau(\alpha) = x_{(r)}, \qquad r = \lceil (n+1)(1-\alpha) \rceil , \tag{1}$$

with $\tau = +\infty$ if $\alpha \leq 0$, $n = 0$, or $r > n$; alerts require $s > \tau$. Frontier utility $u_k(\alpha)$ is the mean recall across attack subtypes. Frontier benign scores determine provisional thresholds used to evaluate recall on labeled validation attacks. Final thresholds are subsequently recalibrated using a separate benign partition. The allocation requires labeled validation attacks and is not a label-free adaptation method. Eligible subtypes require at least 50 validation rows, clients require at least 200 rows across two subtypes, and the 207-point grid spans $[0, 0.05]$ and includes every evaluated budget.

Training data fit the detector and the client normalization, and the test partition supplies the reported FPR and recall. Reusing the frontier sample as final calibration would make the frontier utility estimate and the deployed threshold depend on the same benign records; separating the partitions avoids that direct reuse but does not remove source-order distribution differences.

For each candidate rate, a provisional frontier threshold is applied to the validation attack scores; repeating this across the grid gives one estimated utility curve per client. Once a grid point is chosen, the provisional threshold is discarded. For a fixed benign sample, neighboring nominal rates can select the same order statistic and produce identical alerts. The allocation grid therefore contains nominal choices that may correspond to the same empirical operating point. A difference between selected rates need not produce a difference in calibrated FPR or recall.

Recall is averaged first across eligible attack subtypes within each client and then across clients. A client with more test attack rows therefore does not automatically dominate the macro objective. The minimum-client endpoint uses the smallest of those client-level macro recalls. Macro recall summarizes average client performance, whereas minimum-client recall identifies the weakest client at each operating point. Both must be interpreted alongside held-out benign FPR because a high recall obtained by overspending the alert budget is a different operating point.

3.2 Allocation and calibration

FABRID-IDS reallocates the nominal alert budget by selecting client-specific grid rates subject to

$$\sum_{k=1}^{K} w_k \alpha_k \leq B. \tag{2}$$

Primary N-BaIoT results use $w_k = 1/K$, so nominal and realized federation FPR are both equal-client averages. One-hot mixed-integer programs (HiGHS) maximize mean u_k (Macro) or the mean of the lowest $q = \lceil 0.25K \rceil$ client utilities (Tail) [13]. For $K = 9$ and $q = 3$, Tail maximizes the mean of the three lowest client utilities, unlike conventional empirical CVaR at tail fraction 0.25, which assigns fractional weight to the third client [13]. Minimum-client recall is reported separately. Candidate frontier rates below twice the order-statistic transition are excluded.

Equal-client weighting defines a mean of device FPRs, not the fraction of all benign records flagged across the combined dataset. The latter would weight devices by their available benign row counts, which do not establish comparable collection durations or operational alert volumes. The constraint permits unspent budget: it is an upper bound on allocated targets, not an equality that forces each device to receive a positive rate. After maximizing estimated utility, Macro minimizes nominal expenditure among near-optimal allocations; Tail first uses estimated mean utility as a secondary criterion, then minimizes expenditure. Both use estimated utility variance and a deterministic tie-break afterward. These stages select among allocations meeting the earlier objective within solver tolerances; they do not use final-calibration or test outcomes. Equal-FPR, the principal baseline, assigns the same nominal target to every client and uses the same final-calibration rule. Thus differences from Equal-FPR arise from the rate assignment rather than a different detector or final-calibration sample.

Improving the mean recall of the three weakest clients does not necessarily improve minimum-client recall. This matters when one client's threshold produces no attack detections: gains for the other low-utility clients can raise the Tail objective while minimum-client recall remains zero.

Final thresholds are recalibrated with Eq. (1). The conservative intervention selects the smallest rank satisfying the binomial calibration construction [10]

$$r_{.95} = \min\{r : \Pr[\text{Binomial}(n, 1-\alpha) \leq r - 1] \geq 0.95\}. \tag{3}$$

The search is over ranks $1, \ldots, n$. If none satisfies Eq. (3), the implementation sets $\tau = +\infty$, yielding no alerts; this also applies when $n = 0$ or $\alpha = 0$. Otherwise the threshold is $x_{(r_{.95})}$, with strict $s > \tau$ alerts. The binomial condition bounds the probability, over repeated benign calibration samples, that the selected threshold's population false-positive rate exceeds α; it does not bound the realized rate on an arbitrary fixed test partition. This rank adjustment is a finite-sample sensitivity intervention rather than a deployment guarantee under the source-order split. Its nominal 95% interpretation requires i.i.d. calibration and future benign scores.

The conservative rule moves the threshold rank upward when its binomial criterion requires it, which can reduce false positives at the expense of attack recall. The binomial

calculation concerns repeated sampling under its assumptions; it cannot correct a shift between the final-calibration and test score distributions or repair a client whose attack scores remain below the threshold. We therefore report it as a sensitivity check against the ordinary order statistic, evaluated on the same held-out test records.

The two calibration rules answer different sampling questions. The ordinary order statistic sets an operating threshold from the available benign scores at the chosen nominal rate. The conservative rank additionally accounts for uncertainty in that threshold when calibration samples are repeatedly drawn from a stable benign distribution. Neither rule observes future test outcomes when selecting its rank. Consequently, a test-set FPR above the allocated target is an evaluation result, not evidence that the nominal allocation constraint was violated. Conversely, a test-set FPR below the target does not establish that the same threshold will remain conservative under a changed traffic distribution. Reporting the selected rate, calibration rank, and held-out FPR separately therefore distinguishes the optimizer's decision, its finite-sample implementation, and the behavior measured on the test partition. This distinction applies equally to optimized allocations and Equal-FPR, which share the final-calibration procedure.

With strict alerts, $\alpha < 2/(n+1)$ gives either $r > n$ or a maximum-score threshold and zero calibration alerts; the smallest nonzero empirical FPR is $1/n$ (absent ties). Rates below this transition cannot generate a nonzero empirical calibration FPR under strict thresholding; excluding them prevents the optimizer from assigning nominal budget to grid points that are empirically indistinguishable from a zero-alert calibration point. Ecobee has $n = 1311$ on the final-calibration split, so targets below 0.001524 calibrate to zero alerts. This calculation explains why sufficiently small allocated rates produce no calibration alerts, but it does not explain the subsequent held-out FPR shift.

3.3 Frozen scorer

The primary scorer is the maximum of frozen frontier-reference percentile ranks for autoencoder reconstruction error and the jitter-based temporal feature `HH_jit_L0.1_mean` (f_{75}). The f_{75} candidate and scorer ablations were predeclared for development-only selection. The autoencoder uses [64,16] layers, ten rounds of three local epochs, row-count-weighted averaging, Adam (0.001), batch size 64, MSE, and per-client training-split normalization; ten detector seeds are the paired replication unit.

Percentile transformation places the two components on a common reference scale before taking their maximum; this is part of the fixed scorer and is not retuned per policy. A policy may change recall by moving a threshold, but it cannot change AUROC or AUPRC computed from the shared scores, so ranking diagnostics provide context rather than policy outcomes.

4 Experimental methodology

The primary experiment uses nine N-BaIoT devices and five prespecified budgets from $B = 0.001$ through $B = 0.020$ (Table 1). Macro recall is the mean of client-level macro recalls over eligible attack subtypes; minimum-client recall is their minimum.

N-BaIoT contains benign traffic and BASHLITE and Mirai attack records from commercial IoT devices [14]. The nine devices are natural clients in this study. Their

benign sample sizes are unequal, ranging from 13,113 records for Ecobee to 175,240 for the Philips baby monitor in the retained data. After the source-order split, the final-calibration sample is correspondingly small for Ecobee. This imbalance makes an equal nominal rate a different empirical threshold problem on different devices even before the attack-score distributions are considered.

The budgets are 0.001, 0.0025, 0.005, 0.010, and 0.020. For each seed, the same trained detector and score partitions are used across the policies, so the paired seed is the unit for the prespecified statistical contrasts and test rows within a seed are not treated as independent replicates. Seed-averaged results describe this fixed benchmark and protocol, not behavior on an unseen deployment population.

Each reported federation FPR is computed from the held-out benign FPR of each device with the same equal-client weights as the nominal constraint, which keeps the units aligned with B. The test is not an independent draw from the calibration distribution: the released records are ordered and their traffic features use overlapping windows, and seed replication reflects detector variation rather than resampling of collection periods or devices.

Prespecified contrasts compare Macro with Equal-FPR on macro recall and Tail with Equal-FPR on minimum-client recall using ten paired seeds, exact sign-flip tests, 50,000-resample paired-bootstrap CIs, and Holm correction over each five-budget family. CIC IoT-DIAD provides a 19-client secondary protocol stress test using 57 features and reconstruction error alone. Because the client population, feature representation, and scorer all change, the experiment is not used for cross-dataset mechanism attribution.

Statistical resampling operates at the seed level: each seed contributes one policy difference per budget, and device outcomes are aggregated into the stated endpoint before testing, so test rows never become replicates. The exact sign-flip test changes the signs of the paired differences, giving at most $2^{10} = 1{,}024$ sign assignments; bootstrap intervals resample paired seed differences with replacement; and Holm correction is applied separately to the five budget-wise tests in each contrast. The uncertainty estimates concern variation across trained detectors on the fixed split, not independent collections of IoT traffic. The 95% bootstrap intervals are pointwise for each budget and contrast, not simultaneous intervals over the five budgets. Holm correction applies to the p values within each endpoint family, not to these intervals. The adjusted p values therefore govern multiplicity-controlled claims, while the intervals summarize the paired seed differences.

The secondary dataset is drawn from the device-identification and anomaly-detection study of Rabbani et al. [15]. A fresh dataset-specific detector is trained for this branch; no N-BaIoT checkpoint is transferred. The branch tests whether the nominal-to-realized distinction remains visible under another dataset protocol.

5 Results

5.1 Nominal and realized operating points

At $B = 0.001$, unrestricted Macro and Tail each realize federation FPR 0.00170 ($1.696B$), whereas their conservative variants realize 0.00036 ($0.358B$) and macro recall falls from 0.3778 to 0.2445. These comparisons use the same nominal budget ceiling B,

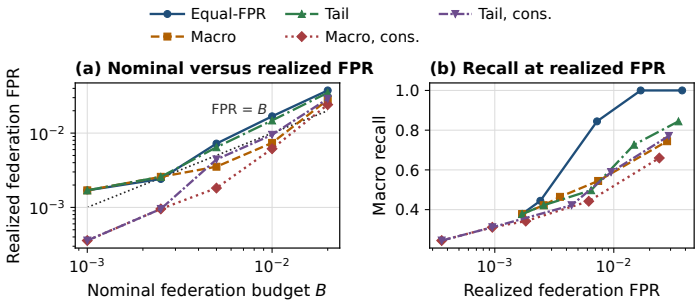


Fig. 1 N-BaIoT ten-seed means: realized FPR versus nominal budget (left) and macro recall versus realized FPR (right). The diagonal marks equal rates.

Table 1 N-BaIoT ten-seed means. Each policy cell reports macro recall; realized federation FPR divided by nominal budget. All FABRID-family cells have zero minimum-client recall.

B	Equal-FPR	Macro	Tail	Macro, cons.	Tail, cons.
0.0010	0.3778; 1.70	0.3778; 1.70	0.3778; 1.70	0.2445; 0.36	0.2445; 0.36
0.0025	0.4445; 0.97	0.4222; 1.03	0.4222; 1.03	0.3111; 0.38	0.3111; 0.38
0.0050	0.8444; 1.45	0.4644; 0.71	0.4977; 1.29	0.3422; 0.36	0.4222; 0.88
0.0100	0.9999; 1.68	0.5444; 0.74	0.7266; 1.48	0.4422; 0.62	0.5889; 0.95
0.0200	1.0000; 1.88	0.7444; 1.40	0.8444; 1.75	0.6600; 1.21	0.7711; 1.45

but neither allocated expenditure nor realized federation FPR is necessarily equal across policies. The optimizer constrains $\sum_k w_k \alpha_k$, whereas evaluation measures the FPR produced after independent local calibration. Consequently, an allocation can satisfy the nominal constraint while exceeding or undershooting the budget after calibration.

The direction of the mismatch changes with the budget and policy. At $B = 0.0025$, Macro and Tail are close to the nominal rate at $1.03B$. Equal-FPR likewise exceeds its nominal budget at several points. Budget overshoot is therefore not exclusive to optimized allocation. Because realized FPR differs across policies, comparisons at the same nominal budget do not establish superiority at a common measured FPR. Figure 1(b) compares macro recall against realized federation FPR.

At $B = 0.005$, Equal-FPR reaches macro recall 0.8444 at a realized FPR of $1.45B$. Macro reaches 0.4644 at $0.71B$, while Tail reaches 0.4977 at $1.29B$. At $B = 0.020$, Equal-FPR reaches recall 1.0000, compared with 0.7444 for Macro and 0.8444 for Tail. The corresponding realized FPR ratios are $1.88B$, $1.40B$, and $1.75B$. Equal-FPR has higher recall at these budgets, but at different measured FPRs.

Allocation is sensitive to the frontier evidence. For each seed and budget, 500 bootstrap replicates resample frontier benign scores and validation attack scores within subtype, then rerun unrestricted Macro allocation; the scorer, final-calibration scores, and test scores stay fixed. Within each seed–client–budget cell, a change means that the selected rate differs from the modal rate among its 500 replicates. Averaging those change frequencies over ten seeds and nine clients gives 77.8, 79.8, 83.9, 83.6, and 84.9% at the five ascending budgets. These are means of 90 cell-level frequencies per budget,

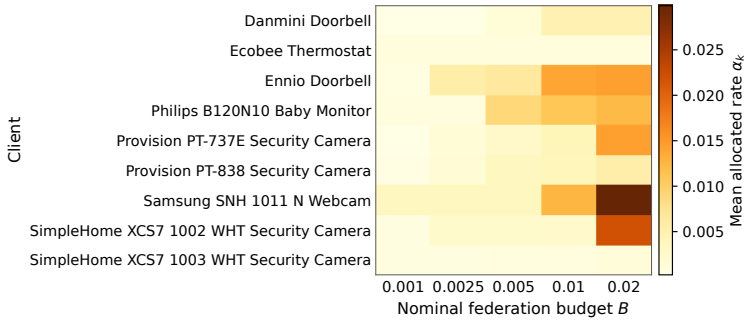


Fig. 2 Original unrestricted Macro rates α_k, averaged across ten N-BaIoT seeds. The heatmap excludes the 500 frontier resamples.

not a count of seeds with any change. They measure sensitivity to resampled allocation evidence, not changes relative to the original allocation or repeated deployments.

Figure 2 separately shows the mean original Macro rate across ten detector seeds for each client and budget; its values do not include the 500 resamples.

5.2 Tail objective and worst-client behavior

Tail optimizes the rounded three-client lower-tail mean, but minimum-client recall is zero for every FABRID policy–budget cell and cannot distinguish unrestricted Macro and Tail. Danmini Doorbell has zero macro attack recall under every evaluated unrestricted-policy operating point; its final-calibration and held-out benign FPR are also zero in all 100 unrestricted-policy–seed–budget cells. With $n = 4955$ benign final-calibration records, its zero-alert transition is $2/(n+1) = 0.000404$; its allocated-rate range crosses this finite-sample transition.

Ecobee illustrates a different calibration/test discrepancy. At $B = 0.020$, Macro assigns Ecobee a mean nominal rate of $\alpha_k = 0.001164$. Its reported final-calibration FPR is zero, while mean held-out benign FPR is 0.006672 and macro recall is 0.6000. Finite calibration resolution explains the zero-alert calibration point, but the experiment cannot separate that effect from distributional differences between the source-order final-calibration and test partitions.

Danmini Doorbell and Ecobee thus differ under local calibration: Danmini has zero attack recall and zero held-out benign FPR, whereas Ecobee records attack detections and held-out benign alerts despite zero final-calibration alerts. The calibration sample sizes (4,955 versus 1,311 benign records) determine the available threshold resolution but cannot alone explain the held-out outcomes, so client behavior is better read from the allocated rate, calibration FPR, test FPR, and attack recall together than from federation-level averages.

5.3 Primary contrasts and ranking ceiling

Macro and Equal-FPR are identical at $B = 0.001$; their -0.0222 difference at $B = 0.0025$ is not statistically distinguishable from zero, and Macro recall is lower at larger budgets. Both Tail and Equal-FPR have zero mean minimum-client recall through

Table 2 Prespecified paired seed-level contrasts ($n = 10$): 50,000 paired-bootstrap CIs and Holm-adjusted exact paired sign-flip p values over the five-budget family.

	Macro − Equal-FPR: macro recall			Tail − Equal-FPR: minimum-client recall		
B	Mean Δ	95% CI	Holm p	Mean Δ	95% CI	Holm p
0.0010	0.0000	[0.0000, 0.0000]	1.0000	0.0000	[0.0000, 0.0000]	1.0000
0.0025	−0.0222	[−0.0556, 0.0111]	0.3594	0.0000	[0.0000, 0.0000]	1.0000
0.0050	−0.3800	[−0.4133, −0.3466]	0.0098	0.0000	[0.0000, 0.0000]	1.0000
0.0100	−0.4555	[−0.5000, −0.4111]	0.0098	−0.9998	[−0.9998, −0.9998]	0.0098
0.0200	−0.2556	[−0.2889, −0.2222]	0.0098	−0.9999	[−0.9999, −0.9999]	0.0098

$B = 0.005$. At $B = 0.010$ and 0.020, Equal-FPR reaches 0.9998 and 0.9999 minimum-client recall while Tail remains at zero (Table 2); Equal-FPR's realized FPR is also $1.68B$ and $1.88B$ at those budgets.

Because allocation changes thresholds rather than anomaly scores, AUROC and AUPRC are policy-invariant by construction. For each detector seed, the implementation pools held-out benign scores across all nine clients and attack-test scores across subtypes, then computes a rank-based AUROC and a trapezoidal precision–recall area. The reported 0.9995 AUROC and 0.9999 AUPRC are means of these pooled values over ten seeds, not of client-level areas, with attack subtypes and clients weighted by held-out row count. From the retained counts and split boundaries, the test pool has 1,041,078 attack and 111,188 benign rows (90.35% attack prevalence), which raises the AUPRC reference level. Because pooling compares scores across clients, neither area establishes strong ranking for every device or subtype, and neither describes performance at a calibrated threshold: FABRID-family minimum-client recall remains zero across all five N-BaIoT budgets.

The paired tests in Table 2 compare policies at equal nominal budgets. The confidence intervals describe paired seed differences on this split, not uncertainty over new devices or time periods. Adjusted p values do not make the realized FPRs equal. A common-measured-FPR comparison would require a separate design and matching rule; lines in Fig. 1 only connect the five tested budgets.

5.4 Secondary protocol stress test

CIC IoT-DIAD provides a secondary protocol stress test of unrestricted Macro allocation with ordinary local order-statistic calibration. Table 3 averages held-out evaluations over ten independently trained detector seeds, with 19 equally weighted physical-device clients and reconstruction-error scoring. The benign splits and attack-validation fraction match N-BaIoT; each seed trains a new detector, and no N-BaIoT weights or score rankings are transferred. Across the five nominal budgets, macro recall ranges from 0.3614 to 0.5016, realized federation FPR from 0.0098 to 0.0311, and minimum-client recall from 0.0000 to 0.0251.

The realized federation FPR also exceeds the nominal budget in this dataset, but different clients, features, and scoring prevent attribution to the same mechanism as in N-BaIoT. Across the five budgets, the secondary realized FPR is 9.80, 4.72, 3.20,

Table 3 CIC IoT-DIAD: unrestricted Macro with ordinary local calibration; ten-seed means over 19 equally weighted clients using reconstruction-error scores.

B	Macro recall	Minimum-client recall	Realized FPR
0.0010	0.3614	0.0000	0.0098
0.0025	0.3970	0.0000	0.0118
0.0050	0.4318	0.0007	0.0160
0.0100	0.4602	0.0089	0.0213
0.0200	0.5016	0.0251	0.0311

2.13, and 1.56 times the respective nominal ceiling. These decreasing ratios do not indicate a decline in absolute false alarms: realized FPR increases from 0.0098 at $B = 0.001$ to 0.0311 at $B = 0.020$. Over the same interval, minimum-client recall reaches only 0.0251. Because the denominator changes with B, the ratio alone can mislead; absolute FPR and recall must be considered together. The secondary table contains unrestricted Macro only, with no Equal-FPR comparator under the same clients and scorer. It therefore supports the observed budget mismatch but cannot establish whether reallocation improves the recall–FPR relationship relative to a shared nominal client rate.

6 Discussion

On N-BaIoT, nominal budget feasibility does not imply realized budget fidelity. Equal-FPR achieves higher macro recall at several budgets, so the evidence does not support uniform superiority of any allocation policy.

Equation (2) constrains the weighted sum of nominal client FPR targets, not the false-positive rates observed after calibration. Unrestricted allocation can exceed the nominal federation budget, while conservative calibration reduces false positives at the cost of attack recall. Neither approach recovers detection for Danmini Doorbell at the evaluated operating points. A comparison of allocation policies must account for the allocated budget, realized federation FPR, macro recall, and worst-client recall.

The optimizer values rates using attack recall measured against thresholds from frontier benign scores. The chosen rates are then translated into new thresholds using independent final-calibration scores, and recall and FPR are measured on held-out scores. The nominal-to-realized mismatch and Ecobee's zero-alert calibration but nonzero test FPR show that these stages can disagree. The data do not isolate whether finite-sample variation, score-distribution differences, or another source produced a particular discrepancy, and source order does not establish a common deployment timeline, so temporal drift is not demonstrated.

Macro and Tail require representative labeled validation attacks to estimate client utility curves, so the findings characterize a post-training allocation protocol rather than a label-free adaptive system.

Recorded solver times across the five budgets range from 95.7 to 358.8 ms for Macro and from 310.3 to 3883.1 ms for Tail, with no recorded fallback. They are modest relative to detector training but not an end-to-end deployment latency: communication, score generation, calibration transfer, and alert handling are excluded.

The primary evaluation covers one hybrid scorer, nine N-BaIoT devices, ten seeds, and five nominal budgets; the secondary evaluation uses 19 CIC IoT-DIAD clients, 57 features, and reconstruction-error scoring. Because clients, features, and scoring all differ, the secondary results show that budget mismatch occurs in both protocols without identifying a common cause, which would require comparisons that vary individual components while holding the others fixed. Privacy, adversarial clients, and deployment overhead are outside scope, and scorer-level interventions intended to address the observed detection failures were not evaluated.

Table 1 also shows why realized FPR must be reported for each policy. Macro and Equal-FPR coincide at $B = 0.001$, but at $B = 0.020$ Equal-FPR attains higher recall at a higher realized FPR ($1.88B$ versus $1.40B$). A recall difference at equal nominal budgets therefore mixes rate assignment with a difference in realized FPR.

The paired tests in Table 2 have limited resolution. The exact sign-flip test is two-sided and enumerates all $2^{10} = 1{,}024$ sign assignments, so the smallest attainable raw p value is $2/2^{10}$, reached only when all ten seed differences share one sign. Multiplying by the five Holm comparisons gives 0.0098, the smallest adjusted value attainable here. Entries of 0.0098 mean that the paired differences have the same sign across all ten seeds; their magnitudes are reported separately in Table 2. At $B = 0.001$, identical policies give $p = 1$ and a zero-width interval. The minimum-client intervals at $B = 0.0100$ and 0.0200 are nearly degenerate because Tail's minimum-client recall is zero in every seed, so the difference follows Equal-FPR's recall, which varies across seeds only beyond the fourth decimal. Holm correction is applied separately to the macro-recall and minimum-client families, so the adjusted values do not account for testing both endpoints.

The conservative rank rule also has a narrower guarantee than its 95% level suggests. The binomial condition applies to each client separately, so with nine clients a union bound gives a joint probability of at least $1 - 9(0.05) = 0.55$ that all nine population FPRs are within their nominal rates, under i.i.d. calibration and future scores. The evaluated procedure makes no federation-level statement.

For practical use, the nominal allocation and measured operating point should be reported together. A site that requires a hard observed FPR ceiling cannot rely on Eq. (2) alone; it would need a calibration and monitoring design justified for its sampling process. The conservative rank adjustment illustrates the trade-off: fewer measured false positives in the tight-budget primary case come with lower mean recall and no recovery of the worst client.

Interpreting allocation gains requires keeping the comparison unit fixed. A mean-recall comparison at one nominal budget evaluates the complete policy, including its calibrated thresholds. It does not isolate utility at a common measured FPR, because the policies may reach different realized operating points. A federation average can improve while one device detects no attacks. The Tail objective addresses average utility within a lower group of clients; minimum-client recall evaluates the single weakest outcome. Reading both endpoints prevents gains in one summary from being interpreted as improved protection for every client. Paired detector seeds hold scorer variation common across policies, but do not replace independent traffic collections or repeated calibration samples from a deployment population.

7 Conclusion

FABRID-IDS allocates client-specific nominal FPR targets under a shared federation budget and recalibrates thresholds locally. Nominal compliance can still produce realized FPR mismatch or zero worst-client recall. Equal-FPR attained higher macro recall at several nominal budgets, and no FABRID policy raised minimum-client recall above zero on N-BaIoT, so the evidence supports the distinction between nominal and realized FPR rather than a performance advantage for reallocation. Limitations include source-order calibration, attack-informed utility, and the nine-device benchmark. Future work should study deployment-ordered calibration, distribution shift, and realized-FPR control under explicit sampling assumptions. The implementation and reproducibility artifacts are available at https://doi.org/10.5281/zenodo.22113172. The authors declare no competing interests.

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